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Participatory Sensing and Digital Twin City: Updating Virtual City Models for Enhanced Risk-Informed Decision-Making

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Abstract: The benefits of a digital twin city have been assessed based on real-time data collected from preinstalled Internet of Things (IoT) sensors (e.g., traffic, energy use, air pollution, water quality) for managing the complex systems of cities, but the sensor-based reality information is likely insufficient to provide dynamic spatiotemporal information about physical vulnerabilities. Understanding cities' current states of physical vulnerability can support city decision makers in analyzing associated potential risk in urban areas for data-driven infrastructure management in extreme weather events. As a step toward creating a digital twin city for effective risk-informed decision-making, this paper proposes a new framework to bring crowdsourced visual data-based reality information into a three-dimensional (3D) virtual city for a model update with interactive and immersive visualization. Unstructured visual data are collected from participatory sensing and analyzed to estimate the geospatial information of vulnerable objects in the distance representing physical vulnerability in cities. The crowdsourced visual data–based reality information of physical vulnerability in a given region is then integrated with a 3D virtual city model, and the updated 3D city model is fed into a computer-aided virtual environment (CAVE) for immersive visualization to enable users to navigate the intersection of reality and virtuality. To test the proposed framework, case studies were conducted on Houston. The outcomes demonstrate that the proposed method has the potential to make the virtual city model live in terms of local vulnerability. The digital twin city building on crowdsourced visual data is expected to contribute to risk-informed decision-making for infrastructure management in cities and help analyze various what-if scenarios in disaster situations with increased visibility of hazard and city interactions. **DOI: 10.1061/(ASCE)ME.1943-5479.0000748.** *© 2020 American Society of Civil Engineers.*

Author keywords: Participatory sensing; Digital twin city; Geospatial localization.

Introduction

The proportion of the world's population who live in cities is increasing rapidly, from 54% in 2014 to an estimate of almost 80% by 2050 (United Nations 2014). As a result, city decision makers must consider how to keep the sustainable and resilient connectivity between humans, infrastructure, and technologies that is essential for community functioning in cities. Decision makers continually struggle with various challenges posed by rapid urbanization and need to become more strategic in responding to these challenges. Many cities are increasingly implementing technological development for a data-driven approach, which can facilitate effective decision-making in complex systems of cities. Recently, many cities have started to create and leverage a virtual replica of a city (i.e., a digital version of a real city) at the intersection of reality and virtuality (Mohammadi and Taylor 2017). This new technology has an important characteristic: real-time data using sensors such as Internet of Things (IoT) devices to bridge between the real world and its virtual counterpart and understand relations among essential constituents of a city, with increased visualization of human–infrastructure–technology interactions, to facilitate decision-making. For example, the city of Rennes, in the Brittany region of France, has carried out the Virtual Rennes project, building a three-dimensional (3D) virtual twin of the actual city to establish a new vision of the city, accommodate the growing population, and design new planning and city management in terms of transportation, energy consumption, and environment (Cooper 2018). A municipality of Amsterdam has also developed a digital twin of the Amsterdam Southeast area for managing energy transition. As a medium of citizen participation, the virtual city enables decision makers to share creative ideas with others in a dynamic way as well as to devise potential solutions in a virtual game environment (Geodan 2018; Oud 2018).

Because the challenges in effective decision-making in such complex systems are likely to be exacerbated in extreme situations, the digital twin city, integrated with real-time data collected from city monitoring systems, has been leveraged to facilitate risk-informed decision-making given the increasing numbers and impacts of disasters. For example, the Virtual Singapore project has created a 3D city model with a variety of data sources, such as geospatial information from government agencies and real-time data from IoT sensors around the city (Dassault Systèmes 2018; Holstein 2018; Virtual Singapore 2018). In this project, the real-time data recorded from various sensors were brought into contextualized 3D city models to understand potential disaster risk, such as electrical network failures, and then analyze evacuation models to optimize evacuation plans and minimize possible structural damage and loss of life. As such, understanding the states of spatiotemporal flux in cities before, during, and after disaster situations

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(e.g., extreme weather events) is effective for decision-making that influences physical, environmental, and social systems of cities in terms of urban risk management.

In spite of the benefits of the digital twin city with real-time data collected from preinstalled IoT sensors for monitoring energy use, air pollution, water quality, noise pollution, and traffic in cities, the sensor-based reality information is not sufficient to monitor the extent to which urban environments are vulnerable to potential natural disasters for both disaster preparedness and hazard mitigation. Although live data obtained from preinstalled city monitoring sensors enable governmental agencies in major cities to monitor associated real-time information in extreme weather events (e.g., current conditions of electric power systems or major transportation-related infrastructure), it is not feasible or straightforward to know the high-resolution contextualized spatiotemporal data about physical vulnerabilities—widely distributed in cities—that are likely to be affected by extreme weather events. Physical vulnerabilities in cities (vulnerable objects or potential wind-borne debris around critical infrastructure, neighborhood physical disorder, etc.) potentially lead to primary or secondary hazards in extreme weather events, which may result in economic, environmental, and social damage to communities and neighboring interdependent critical infrastructure systems. A readily available source of such data would likely allow better identification and understanding of current states of physical vulnerabilities, which would help predict future behaviors on which such physical vulnerabilities would have negative impact during forecasted extreme weather events, affecting communities and neighboring critical infrastructure.

To address challenges in preinstalled sensor-based approaches for in-depth city monitoring, participatory sensing has recently emerged as one way to document the state of built environments for disaster management with little incremental cost (Goldman et al. 2009; Christin et al. 2011; Xue et al. 2018). Although prior studies provide encouraging proof-of-concept supporting the effectiveness of crowdsourced data in the context of disaster situations, they have limitations: (1) they have mostly concentrated on social media messages in the form of text, and in a few studies using photos; app users are asked to manually input detailed information on current damage states shown in photos and associated risks; and (2) they have primarily attempted to directly link "georeferenced social media activity" to "disaster areas" with the assumption that problems occur at the location where original photos are taken. Consequently, objects in the distance of photos that have the potential for causing cascading risks to cities (e.g., potential wind-borne debris beside roads or critical facilities, vulnerable cranes on urban construction sites surrounded by many utility poles) have not been fully investigated in participatory sensing. Because there are likely to be many cases in which people report vulnerable objects in the distance before disaster situations, it is essential to reliably localize such distant objects that may be vulnerable to extreme weather events and map their spatiotemporal information (their geospatial location at the time of the data collector's report). This will help understand the potential risk of physical vulnerabilities in urban areas, which can support effective risk-informed decision-making.

To overcome these challenges, and as a step toward creating a digital twin city for effective risk-informed decision-making, this paper aims to propose a new participatory-sensing-to-digital twin city framework. The proposed method leverages unstructured crowdsourced visual data to identify and localize vulnerable objects in urban areas, and the resulting spatiotemporal information of physical vulnerability is integrated with a 3D virtual city model. The updated 3D city model, reflecting the spatiotemporal information of physical vulnerabilities, is then fed into an interactive computer-aided virtual environment (CAVE) for interactive visualization of participatory sensing–based reality information in a virtual environment.

The primary contributions of this paper can be summarized as follows: (1) leveraging unstructured crowdsourced visual data to better understand spatiotemporal information of physical vulnerabilities in cities, enhancing modeling accuracy of cities by reducing the gap between a real city and its digital replica; and (2) bringing an additional context dimension into current IoT-based digital twin city environments by providing an interactive visualization of spatiotemporal information of physical vulnerabilities, enabling users to interact with the updated 3D city model in the CAVE at the intersection of humans–infrastructure–technology. This study contributes to the body of knowledge of data-driven infrastructure management by providing a novel participatory sensing-to-digital twin city framework. This framework can help decision makers better understand the as-is condition of city infrastructure in terms of vulnerability to anticipated extreme weather events, which can greatly assist data-driven infrastructure disaster management.

Eventually, the proposed digital twin city models can be used to simulate various what-if scenarios based on contextualized geospatial and semantic information. This approach can become a basis of effective risk-informed decision-making for infrastructure management in terms of disaster response and recovery, as well as disaster preparedness and hazard mitigation against potential extreme weather conditions. In the following sections, previous works on a digital twin city based on sensor data, participatory sensing for disaster management, and distant object localization are reviewed, and their challenges are discussed. Then, the proposed methods are presented in detail. Finally, experimental results of case studies on Houston, the implications of this research, and open research challenges are discussed.

Related Works

Digital Twin City and Participatory Sensing in the Context of Disaster Management

The term "digital twin" can broadly be defined as a creation of a dynamic virtual representation of a physical system, along with connectivity and analytical capabilities using IoT (National Infrastructure Commission 2017). In particular, in regard to urban risk management, a digital twin city informed by sensor-based real-time data has the potential to better understand the spatiotemporal information in cities, supporting effective risk-informed decision-making. For instance, a gas leak was simulated in a contextualized 3D city model with reality data to identify potentially affected areas and inform city dwellers in those areas of escape routes (Coletta et al. 2019). As another example, Singapore's security forces could use a digital copy of Singapore to simulate a terror threat at a sports stadium to prepare appropriate reactions (Geddie and Aravindan 2018). Similarly, the city of Newcastle has performed a flood simulation in a computer replica based on abundant real-time data from sensors installed all around the city (White 2018).

Although live data obtained from monitoring sensors have the significant potential to improve risk-informed decision-making, there is still a gap in the current digital twin city, as it is not trivial to identify and understand physical vulnerability by using preinstalled sensors. Critical to disaster response efforts relevant to urban environments during or after an incident is gathering and making sense of information. Decision makers in most regions, especially major cities, already have systems to gather near-real-time information about major critical infrastructure (e.g., which bridges are

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likely flooded and which need to be closed for further inspection). However, inspecting numerous vulnerable objects in cities is time-consuming and labor-intensive. Participatory sensing, enabling city dwellers to report disaster-related information using mobile devices, has attracted considerable interest as an alternative tool for data collection because of its scalability over large urban areas. Several studies (including Eilander et al. 2016; Fohringer et al. 2015; Jongman et al. 2015; Muralidharan et al. 2011; Restrepo-Estrada et al. 2018; Sun et al. 2016; Wang et al. 2018) have been conducted on participatory sensing with crowdsourced data to document the state of built environments during or after disasters [e.g., rainfall runoff estimation and flood forecasting with social media messages (Restrepo-Estrada et al. 2018), near-real-time flood map based on social media messages (Eilander et al. 2016), urban flood monitoring with photos collected from a crowdsourcing app along with Twitter messages (Wang et al. 2018), rapid inundation mapping from photos posted in Twitter (Fohringer et al. 2015)]. The prior works assumed that problems occur at the reporting location (where photos are taken) and thus simply linked "georeferenced social media activity" to "disaster areas" even though in disaster situations, people likely report vulnerable objects in the distance. Such distant objects at risk typically have the potential for causing primary or secondary hazards in cities and require special attention for disaster preparedness and hazard mitigation. Thus, for effective risk-informed decision-making through participatory sensing, it is necessary to understand the spatiotemporal information of distant objects vulnerable to extreme events.

Geospatial Localization of Distant Objects

Geospatial localization denotes the process of localizing objects or areas of interest in geographic coordinate systems. In geospatial localization, the spatial relationship between a sensing device and an object of interest should be estimated in connection with an absolute location. Generally, the distance between a sensing device and an object of interest is directly measured by using active sensors such as laser scanners. Without active sensors, one alternative to measure the distance is to infer the spatial relationship of an object and a sensing device based on 2D images containing the object and the geographic location of the sensing device. For this purpose, intrinsic and extrinsic parameters of cameras are identified to estimate the 3D geometric information of 2D image contents in connection with the real world (Piasco et al. 2018).

Previous studies on computer vision have presented various methods to localize distant objects or indoor locations in construction sites and built environments (such as Ha et al. 2018; Ham and Yoon 2018). However, their applicability could be limited in urban areas because of the necessity of supplementary information (e.g., building information models) or a certain monitoring condition (e.g., additional cameras required to infer the spatial relationship). In recent years, geospatial localization methods for urban environments have been presented using smartphone sensor data. Manweiler et al. (2012) localized a distant object using the spatial relationship of 3D point clouds reconstructed from a Structure-from-Motion, a global positioning system (GPS) location, and a compass bearing. Ouyang et al. (2013) and Chen et al. (2016) used crowdsourcing-based approaches to localize an event in outdoor environments using shooting directions and GPS locations of multiple users. However, the prior works yielded unstable localization accuracies [e.g., distance errors ranging from 5 to 150 m (Manweiler et al. 2012)] that would be insufficient to differentiate objects that are close to each other in urban environments, so the methods still have room for improvement.

Methodology

Fig. 1 illustrates an overview of the proposed participatory-sensing-to-digital twin city framework to support effective risk-informed decision-making based on an understanding of spatiotemporal information of physical vulnerabilities in cities. In this paper, vulnerable objects are defined as physical objects that likely will lead to

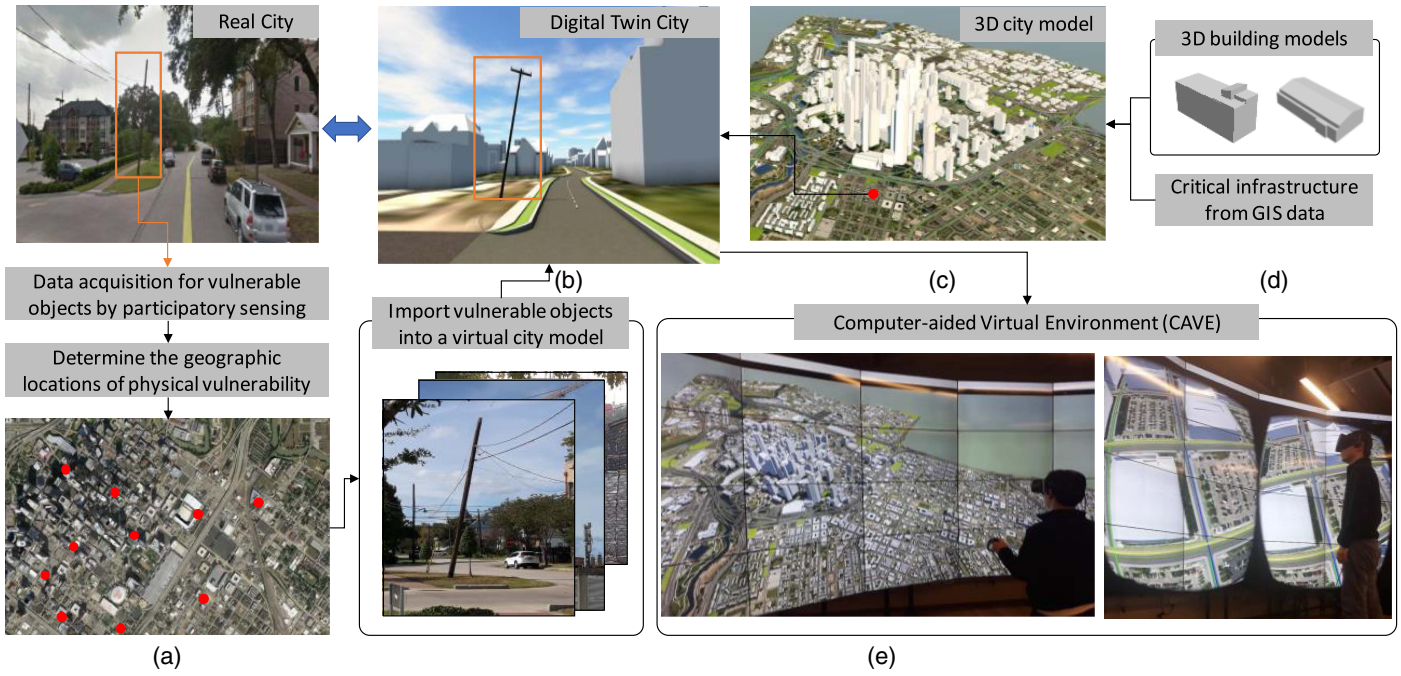


Fig. 1. Overview of the proposed participatory-sensing-to-digital twin city framework. [(a) Image capture: Dec 2018 © 2019 Google; (e) Imagery © 2019 Google, Map data © 2019.]

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primary or secondary hazards in extreme weather events (e.g., high wind), which may result in economic, environmental, and social damage to community functioning and neighboring interdependent critical infrastructure systems. These hazards range from structural defects (e.g., leaning poles) to elements at risk (e.g., potential windborne debris). Physical vulnerability denotes the degree of damage/loss to a given vulnerable object, or set of vulnerable objects, within the area affected by extreme weather events. Crowdsourced data representing such vulnerable objects in given areas will continually update virtual city models through the proposed method, complementing real-time data collected from IoT sensors to better characterize the physical vulnerabilities of given areas and ultimately lead to building a digital twin city in the context of urban vulnerability for enhancing risk-informed decision-making.

Localizing Distant Objects Vulnerable to Potential Natural Disasters Using Crowdsourced Visual Data

We first identify the geospatial location of potentially vulnerable objects in the distance based on mobile sensor data, including the geographic location of a data collector, a compass bearing toward a target object, and a target object image using commodity-level smartphones. With this information, we formulate lines connecting data collectors to a distant object of interest and then determine the location of the distant object based on the intersection points of those lines. With citizen participation, data collection for distant objects is performed through the two steps as shown in Fig. 2.

For data collection, it is assumed that participants take a photograph of a distant object and point the upper part of smartphone toward the distant object while leveling it to the ground to report a compass bearing in a dedicated mobile application program. Through the data filtering process, we verify whether the photo collected involves a target object of interest. In our experiments, it was observed that the data collection generally takes about 10 s per instance, which does not result in significant resistance or discomfort to participants and thus has the potential to be leveraged at scale.

In participatory sensing–based geospatial localization, the data collected are preprocessed to project a geographic location represented in spherical coordinates (latitude, longitude, and altitude) onto the Universal Transverse Mercator (UTM) coordinate system. As a result, the geographic location is represented as 2D Cartesian coordinates—easting and northing—without altitude information. This projection facilitates the localization of a distant object in the proposed method, because the intersection point of multiple line equations can be more efficiently found in a 2D space than localization in a 3D space. Moreover, the data accuracy of altitude is not typically reliable in urban areas because of the interference of nearby trees and buildings, so to improve localization accuracy, it is better not to include such unreliable information in the geospatial localization process. If two or more data points for the same distant object are collected, the geospatial localization method can promptly identify its location. However, because multiple observations for different distant objects can be reported simultaneously in a certain urban area, it is necessary to categorize data points by each distant object. To that end, we use a two-stage filtering process, consisting of location-based clustering and object presence–based filtering.

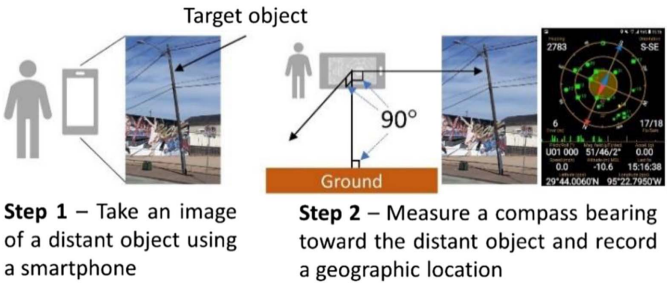


Fig. 2. Data collection process by a data collector.

Location-based clustering divides entire data points into separate clusters based on the proximity between contiguous data points. With this process, data points in a cluster are likely to correspond to the same object of interest, thereby having the potential to improve the reliability of the localization. Data points are categorized into separate clusters building on mean-shift clustering (Cheng 1995) by optimizing a parameter of window size. With the specified window size, data points can be grouped into various numbers of clusters in accordance with their locations and proximity to adjacent data points. Nonetheless, a cluster may still involve irrelevant data points with respect to the distant object of that cluster. For example, although most data points in one cluster correspond to a vulnerable tower crane, some of them may correspond to a leaning utility pole nearby. In such cases, localization accuracy may be significantly degraded because of an erroneous compass bearing of irrelevant data points.

To prevent such undesirable cases, the second filtering stage, object presence–based filtering, builds on a region-based fully convolutional network (R-FCN) (Dai et al. 2016) for object detection to filter out data sets involving image frames without a target object. R-FCN is a kind of deep convolutional network to detect objects in the form of bounding boxes in images, based on a feature extraction network, a region proposal network, and a position-sensitive region of interest (RoI) pooling layers. To robustly detect objects of interest in images, the 50-layer residual network (He et al. 2016), which has been validated as one of the best convolutional networks to extract hierarchical visual features, is leveraged as a feature extractor in R-FCN. Based on extracted visual feature maps, the following convolutional layers—a region proposal network and a position-sensitive RoI pooling layers—identify whether they contain a target object to localize. When an image does not involve a target object, the geographic data and compass bearing associated with that image frame are discarded in the process, because it is likely to have wrong information with respect to an object of interest.

Before geospatial localization, it is necessary to consider the magnetic declination at different locations to adjust compass bearings. The compass bearings are measured by referencing the Earth's magnetic north as indicated by a magnetic sensor; however, because the north of geographic coordinate systems is based on true north, the angle difference (magnetic declination) between magnetic north and true north causes a compass bearing error. To adjust for such deviation, the International Geomagnetic Reference Field Model (National Centers for Environmental Information 2018) finds a degree of magnetic declination at a certain location.

Once the compass bearing is adjusted, a line equation is formulated on each data point in the UTM coordinate system, connecting a data collector's location and a corresponding target object. In cities, it is likely to have multiple observations by citizens for a single object of interest. In that case, hundreds or thousands of possible intersection points among multiple lines could exist due to errors in data collection and inaccurate sensors. After converting multiple line equations into a linear system of equations in a matrix form, a unique solution for the location of a distant object (i.e., intersection of multiple lines) is derived, building on the Moore–Penrose generalized inverse approach (Serre 2010), which builds on a pseudo-inverse matrix of a singular or nonsquare matrix.

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Although the unique location of a distant object can be obtained by this inverse approach with all data points in a cluster, errors of geospatial localization may vary significantly depending on the accuracy of collected data points. To determine the accurate location of a distant object, the 2D distribution of possible locations of a distant object is examined. To this end, the possible locations are estimated by selecting n number of data points (n is the same as or less than the total number of data points), formulating k number of line equations (k is the same as or less than n), and estimating a unique intersection point in each case. With possible combinations of n and k, numerous locations of distant objects are estimated. The final location of a distant object is determined through mean-shift clustering on the estimated locations. A cluster with the largest number of data points is selected, and its cluster center location is regarded as the final location of a distant object.

3D Virtual City Modeling, Its Update Based on Participatory Sensing–Driven Reality Information of Physical Vulnerabilities, and Interactive Visualization in a CAVE

Once the spatiotemporal information of potentially vulnerable objects in the distance is identified in cities, we automatically stream unstructured crowdsourced visual data–driven reality information into a 3D city model with the increased visibility toward an updated 3D virtual city model (Fig. 3).

A 3D virtual city model refers to a digital representation of urban areas that includes 2D or 3D geometry of earth surfaces, buildings, vegetation, and infrastructure, as well as related objects belonging to cities (Biljecki et al. 2015). In this paper, the 3D virtual city model consists of 3D building models, a 2D geographic information system (GIS) map, and critical infrastructure in the form of a set of geometric elements such as points, lines, and polygons in a virtual environment. The geometry of 3D buildings is modeled in the extensible markup language (XML) schema to describe the surfaces, boundaries, and differentiated roof structures of the buildings based on geometric meshes using vertices, edges, and faces. The XML schema is made up of a set of media assets that are individual entities of digital media, including synthetic and photographic bitmaps, 3D mesh models, shaders, animation data, and video samples. In the XML schema, the level of detail (LoD) (i.e., the degree of abstraction of the real world according to computational and economical facets as well as users' needs) must be considered when describing the details of the geometry of building models. In our case studies, we leveraged 11,457 geometric 3D building models to describe parts of Houston in a virtual space, with the level of Detail 2 (LoD2) having detailed shapes of buildings and differentiated roof structures for interactive visualization in a virtual environment.

To map the 3D building models onto a 2D GIS map, the spherical coordinate in the XML schema is used to specify the location of each 3D building model on a 2D GIS map by pairing with the corresponding element ID in the XML schema. A 2D GIS map, one of the components of a 3D virtual city model, is made up of 2D aerial images that represent a flat surface to show the size and shape of areas, locations of features, and distances between multiple points in space with geospatial and semantic information. In this research, the 2D GIS map, consisting of the geospatial data of infrastructures such as highways and railways from the OpenStreetMap (OpenStreetMap 2019), is overlaid with high-resolution satellite imagery, which represents the semantic information for detailed characterization of geometric objects in given regions. OpenStreetMap is an open data repository to provide map images and all underlying map data built by a community of contributors who maintain and update data about roads, railways, buildings, and so on. It includes not only boundary conditions to determine the size of a 3D city model, but also geometric networks involving the terrain, roads, and railways in the areas of interest to model a series of interconnected infrastructures. In this paper, satellite imagery from Microsoft Bing Maps is draped over the surface of a 2D GIS map based on the spatial data of OpenStreetMap and National Elevation Dataset (NED) (United States Geological Survey 2019), such as digital elevation and geographic information of roads and railways. The area on which 3D building models were mapped (parts of Downtown Houston and midtown, museum park, and Texas medical center) was 26.2 km^2.

Critical infrastructure refers to assets, systems, and networks, whether physical or virtual, to maintain normal conditions in daily life (Interagency Security Committee 2015) and provide essential functions to continually operate communities. Critical infrastructure could be damaged or disrupted by failures of elements at risk in potential natural disasters, resulting in the spread of negative impacts on other communities or secondary hazards that cause collapses in other interconnected critical infrastructure systems (Rehak et al. 2018). Thus, to avoid catastrophic collapses of the whole system caused by cascading effects, it is important to understand

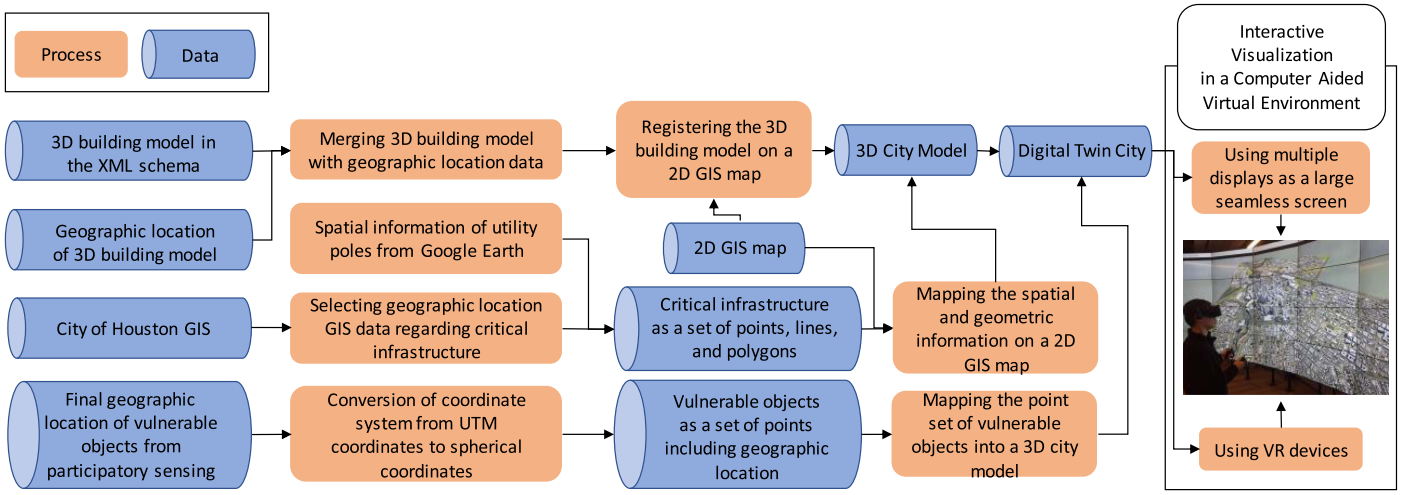


Fig. 3. Overview of the data and process toward an updated 3D virtual model of Houston. XML, extensible markup language.

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spatiotemporal relations between failures of vulnerable objects and neighboring interdependent critical infrastructure.

We have discussed the information of geospatial locations of vulnerable objects from participatory sensing; next, we focus on the geospatial information of critical infrastructure (transportation systems, electric power transmission, water supply, medical care facilities, and emergency facilities). As a proof-of-concept, we mapped critical infrastructure (e.g., bus routes, metro railway, stations, utility poles, schools, hospitals, and storm sewers) around vulnerable objects recognized from participatory sensing in Houston. To gather such critical infrastructure information, we leveraged the GIS data from the City of Houston Geographic Information System (COHGIS) that provides spatial and semantic information in Houston. Here, the critical infrastructure is modeled in the Keyhole Markup Language (KML) schema, which is based on XML standard, for expressing geographic annotation and visualization. The KML schema, using a tag-based structure with elements and attributes, typically displays geographic data by using placemarks, ground overlays, paths, and polygons in web-based 2D or 3D platforms. The KML schema specifies a set of elements and attributes (e.g., placemark, path, polygon, description, ground overlay) with spherical coordinates that can be fed into a 2D GIS map. In this paper, the geographic location of critical infrastructure in the COHGIS is registered into a 2D GIS map in the form of vector data (i.e., point, line string, or polygon) or a 3D mesh model. In addition to the COHGIS, the spatial information of utility poles was captured at street level from a birds-eye perspective in Google Earth. Fig. 4 illustrates parts of the KML schema and shows an example of mapping the geographic location data of critical infrastructure and vulnerable objects on a 2D GIS map.

The geographic location of vulnerable objects is calculated from unstructured crowdsourced visual data, and then the resulting spatiotemporal information is used for a model update to bring an additional context dimension into a 3D virtual city model. As a proof-of-concept, we mapped vulnerable objects recognized from participatory sensing in Houston (e.g., leaning utility poles, broken fences, vulnerable high-rise cranes, potential wind-borne debris such as temporary signs) to better explore spatial information of vulnerable objects and understand the impact of their failures on neighboring critical infrastructure. The outcomes of the geospatial localization are a set of points representing the estimated geospatial locations of vulnerable objects in 2D Cartesian coordinates within the UTM zone. The 2D Cartesian coordinates (easting/northing in meters) of the geospatial locations of physical vulnerability are converted into the spherical coordinate (latitude/longitude in degrees) to import an additional context dimension (in the form of vector data or 3D mesh models) into a 3D virtual city model. After conversion of the coordinate system, the set of geospatial localization results of vulnerable objects in the KML schema is fed into a 3D virtual city for a model update. In other words, we update a 3D virtual city model based on a set of vulnerable objects recognized from unstructured crowdsourced visual data analytics. Fig. 5 illustrates an example of streaming vulnerable objects as a set of points into a virtual city model.

A CAVE is a dedicated system to create and navigate a virtual world in which users can walk through and be immersed in 3D models rather than just looking on a screen or a 2D drawing. In this paper, case studies were conducted on visualizing and exploring the updated digital twin city of Houston in the CAVE facility at Texas A&M University. The system is composed of a 180° curved wall of 36 monitors in a 9-by-4 configuration to make a large seamless display and a computer with a quad-core 3.7 GHz CPU, 16 GB RAM, and a 8 GB graphics processing unit (GPU). The digital twin city of Houston was fed into the CAVE to integrate real-virtual states of spatiotemporal information in urban environments, which helps users interact with the spatial information of vulnerable objects and neighboring critical infrastructures. Here, the proposed framework leverages the real-world data, including visual data collected from smartphones of city dwellers through the server, and spatial and semantic information of critical infrastructures in Houston from the open database (COHGIS). The KML schema containing (1) geospatial information of vulnerable objects obtained from the proposed localization; and (2) geospatial and semantic information of critical infrastructures provided by COHGIS, and the XML

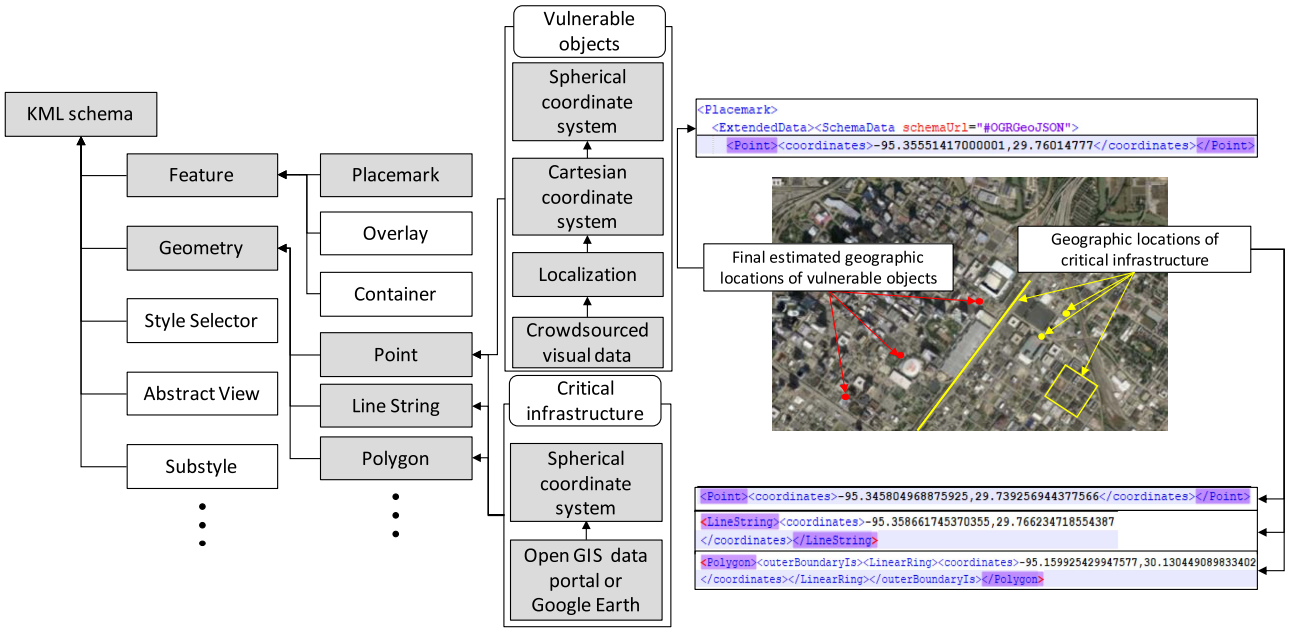


Fig. 4. Logical structure of the elements in the KML schema and an example of mapping critical infrastructure and vulnerable objects. (Imagery ©2019 Google, Map data ©2019.)

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Fig. 5. Example of a vulnerable object recognized through participatory sensing and the localization: (a) an image from Google Earth (without vulnerable condition) (Image Capture: Dec 2018 ©2019 Google); (b) an example of crowdsourced visual data involving a vulnerable object (during our data collection); and (c) crowdsourced visual sensing-driven localization in a city model.

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schema involving geospatial information of urban structures such as building models, consist of various types of data such as source types (vector, raster, 3D model, etc.) and feature types (building, railway, etc.). For example, the KML schema containing the geospatial information of a vulnerable object such as a leaning pole is converted into vector source type and connected with a 3D model representing the vulnerable object for mapping into the 2D GIS map. Similarly, the XML schema describing building models within the boundary of our Houston model is converted into 3D model source type and building feature type. Then, they are placed into the 2D GIS map based on the corresponding geospatial information. By this process, the 3D models of objects of interest are registered into the CAVE for immersive visualization.

In the CAVE, the real-world data (i.e., geotagged visual data collected through participatory sensing and spatial and semantic information of critical infrastructures) is retrieved by interacting with the updated city model in a virtual environment. To retrieve the real-world data in the CAVE, users navigate the city models, which involve 3D models and vector models of critical infrastructures and vulnerable objects mapped by the updated real-world data. Virtual reality (VR) devices such as remote touch controllers help move and rotate a first-person view in the virtual environment to locate the real-world data that users seek. To operate reality-virtuality integration in the CAVE, we leveraged two large display modes that compare the real world with the corresponding virtual city model at the same viewpoint and immersive and interactive VR display in the CAVE, including a VR headset to move and rotate a first-person view for users, a remote touch controller composed of a pair of handheld units to interact and navigate the virtual environment, and a sensor used for optically tracking the position of the VR headset as well as the remote controller. Figs. 6(a and b) illustrate real-world data from Google Street view in Downtown Houston (without vulnerable conditions) and a high-rise crane recognized through participatory sensing (during data collection). Figs. 6(c and d) show a large-scale display of the updated virtual city model and its immersive display in the CAVE.

Experimental Results and Discussion

For evaluation, case studies were conducted in Houston. Vulnerable objects were randomly selected by data collectors. Before the experiments, participants were instructed to be aware of the definition of vulnerable objects in this paper and the data collection process (i.e., how to take photos involving target objects and record the direction information using a dedicated smartphone application while pointing the smartphone toward the objects). In our experiments, as a proof-of-concept, 135 photos were collected from five participants using smartphones (the data collection at scale is discussed in Conclusions), and the collected data involve 11 objects in six categories ranging from structural defects (e.g., leaning poles) to elements at risk (e.g., potential wind-borne debris). In our experiments, the period during which visual data containing the objects of interest were captured represents the temporal dimension. The participant-to-object distances were calculated based on the location where each photo was taken for each object and the ground-truth location of the object. Measured compass bearings were adjusted through the magnetic declination at each location based on the International Geomagnetic Reference Field Model.

Table 1 shows the experimental results obtained by averaging 150 localization experiments for each urban object. The window sizes of the mean-shift clustering were empirically set to be 20 and 10 m at preprocessing and postprocessing, respectively. For each localization, the average computation time was 0.0316 s with the computer having an Intel Core i7-8700 CPU. In the proposed approach, it is technically possible that the localization of a

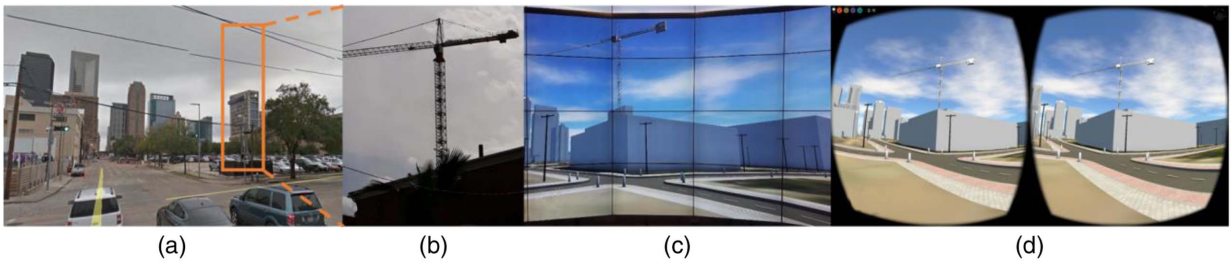


Fig. 6. Example of vulnerable objects in the updated virtual model of Houston compared with the real world at the same viewpoint. [(a) Image capture: Dec 2018 ©2019 Google.]

Table 1. Geospatial localization results of our case studies in Houston

Target object	Data collections (n)	Average participant-to-object distance (m)	Average distance error (m)	Standard deviation of distance error (m)	Final estimated location in UTM system
#1	20	38.1	7.80	2.10	272231.15, 3294527.7
#2	16	26.8	2.94	0.82	272230.02, 3294405.4
#3	9	23.8	11.47	0.58	270055.06, 3293108.6
#4	9	22.2	3.76	0.86	270170.45, 3295169.9
#5	18	31.3	4.01	1.18	272382.8, 3294447.2
#6	12	28.0	3.02	1.53	270071.33, 3293381.2
#7	12	38.7	2.53	0.83	270348.26, 3295069.5
#8	7	24.5	3.00	0.65	270161.45, 3295048.3
#9	14	20.0	4.19	1.23	272178.94, 3293463.7
#10	10	22.8	4.61	1.56	272280.87, 3293605.8
#11	8	18.6	2.11	0.18	272231.46, 3293609.9

vulnerable object is continually updated with additional visual data containing the same object. In this case, only the latest (temporal) version of the localized vulnerability will be modeled and visualized in the virtual environment, as the old one will be replaced with the updated one. Nonetheless, this is a rare case (dynamically changing locations of an object), because we mainly focus on vulnerable objects at the disaster preparedness stage, not their failure scenarios during disaster situations.

In our experiments, it was observed that target object #3, with nine data collections at an average distance of 23.8 m, had an average error of 11.47 m, the largest error. The main reason for such a huge error is the type of target object for localization (broken fences, in this case). When localizing such objects involving large surface areas, localization performance is often degraded because of unreliable compass bearings: it is not trivial to determine a single point of interest for such objects (in this case, determining a single point representing broken fences) involving compass bearing measurements.

To address issues such as this, data collection should be improved by instructing participants in more detail on exactly how to point the smartphone at the object of interest. If the shape of a target object is a point, it can be easy to obtain exact direction information, because there is only one way to orient the smartphone to record direction information. However, most target objects are not in the form of a point, which may lead to collecting inconsistent direction information for target objects, even though data collectors look at the same target from the same position. Thus, for better localization, participants need detailed instruction in recording consistent compass bearings toward target objects.

In our case studies, geospatial localization using unstructured visual data demonstrated the potential for localizing distant objects reported by citizens' smartphones in urban areas. To the best of the authors' knowledge, it is hard to find tolerable error ranges regarding the geographic location of distant objects that can be universally applied. Rather, the proposed method was compared with prior works on localizing distant objects using the sensors embedded in smartphones. Prior works (Chen et al. 2016; Qin et al. 2015; Ouyang et al. 2013; Manweiler et al. 2012) have focused on similar applications: finding the geographic location of distant objects using smartphones (e.g., develop a reporting system to tag objects in the augmented reality (AR) system so that users can interact and communicate with others in the virtual environment). Compared with the performance of such prior studies [e.g., distance errors of about 5–150 m when the distance to the object is about 28–160 m (Manweiler et al. 2012); distance errors of about 3–19 m when the distance to the object is about 20–100 m (Qin et al. 2015); distance errors of about 8–35 m when the distance to the object is about 2–50 m (Chen et al. 2016); distance errors of 9–37 m in the same experimental conditions (Ouyang et al. 2013)], the experimental results of our case study indicate that the proposed framework produces a reasonable localization performance, recording average distance errors of 4.49 m when the distances to the objects at risk were 18.6–38.7 m. The main reason for the slight performance improvement in localization may be that the mean-shift clustering on hundreds of estimated locations with errors removes outliers and helps determine the reliable final location of each target object.

The purpose of geospatial localization in this paper is to help site inspectors to better identify vulnerable objects based on reported locations in the disaster preparedness stage. Regarding the benefits of the proposed framework, the results of the comparison with prior works show that the proposed localization method has the potential to reduce errors when finding the geographic location of distant objects. In this context, in the authors' opinion, the localization error range seems to be acceptable for identifying vulnerable objects (when considering human range of sight) and taking appropriate actions (or establishing action plans) to prevent extreme weather events from causing catastrophic damage to given regions. Despite the significant potential of localizing vulnerable objects for risk-informed decision making, it is noted that localizing distant objects using only the sensors in smartphones, without dedicated ranging sensors, is still challenging, which is an open research problem.

In the matter of transportation-related infrastructure that can impact the daily lives of commuters, businesses, and visitors, as well as the ability of products to move more quickly, cheaply, and efficiently in urban environments, the city of Houston launched the Houston 311 Service in August 2001, a web system to post city emergency or nonemergency information recognized from citizens' participation such as phone calls, an online website, and smartphone applications, to make effective risk-informed decision-making. Here, city emergency information such as geospatial locations of vulnerable objects that may have negative impacts on city transportation systems are classified by municipal agents and scripted in the web system. As a result, reporting of vulnerable objects associated with city transportation systems is shown in a 2D map as a dot containing reporting date, type of complaint, address information, status, and departments that are responsible for solving the issue. In the process of mapping emergency-related information on a 2D map through the city of Houston 311 service, governmental or municipal agents require accurate geographic locations of vulnerable objects; however, the geospatial information recognized from citizens' reporting via phone call or online texting is sometimes not enough to determine the location of vulnerable objects, because most city dwellers rely on a street address to explain an issue location to agents. Furthermore, if city dwellers are at a distance from vulnerable objects, it is not trivial to recognize the

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accurate locations of the distant objects, because many buildings and structures likely obscure the view of distant objects.

To supplement such insufficient information for mapping vulnerable objects, individual government and municipal agencies may need to scrutinize reported issue areas to ensure the geospatial information of vulnerable objects, which would be time-consuming and labor-intensive, especially in the context of disaster situations. To address such challenges, and as a step toward understanding robust spatiotemporal information of vulnerable objects in real time, the proposed method identifies images that include vulnerable objects from unstructured crowdsourced visual data, automatically estimates the geospatial location of such vulnerable objects without dedicated investigation, and maps them into a 3D virtual city model. The proposed method can facilitate participatory sensing, because city dwellers do not need to report the issue locations explicitly; rather, they just take a photo of potentially vulnerable objects in the distance. Thus, the proposed method has the potential to enhance the localization of vulnerable objects compared with the web-based reporting system via phone call or email.

Although it has significant potential to benefit disaster risk management, the proposed method cannot be a single solution to disaster risk management, because the geospatial information of vulnerable objects would not be enough for effective risk-informed decision-making. Thus, it is noted that the details about each issue provided by citizens would complement the proposed framework. Fig. 7 illustrates a process of mapping vulnerable objects through two ways of data collection: citizens' reporting via phone call or online request and the proposed participatory sensing from unstructured crowdsourced visual data.

We selected a vulnerable object (a leaning utility pole) located at State Street between Sabine Street and Trinity Street (object #7) and applied the two different ways for localization and mapping. We observed an error of around 39.66 m (the deviation between the actual location of the object and the mapping location in the web-based system) in the Houston 311 because the system relies on the street address obtained from citizens' manual input for localization and mapping. On the other hand, with the proposed method, the localization error was 2.53 m for the same object, demonstrating the enhanced precision of localizing vulnerable objects. Fig. 8 compares localization errors through the two different approaches for object #7.

Individual government and municipal agencies for security of supply, reliable operation, and optimal management of energy resources implement multirisk assessments for critical infrastructure to improve the resilience of electric networks with respect to diverse facets of the impact of multiple natural hazards. Here, it is important to identify cascading effects on critical infrastructure incurred by failures of vulnerable objects around critical infrastructure, which would be a significant opportunity to improve disaster preparedness and mitigate the effects of disasters. For example, high-rise cranes in urban construction sites surrounded by utility poles could collapse in extreme wind events, and such failures

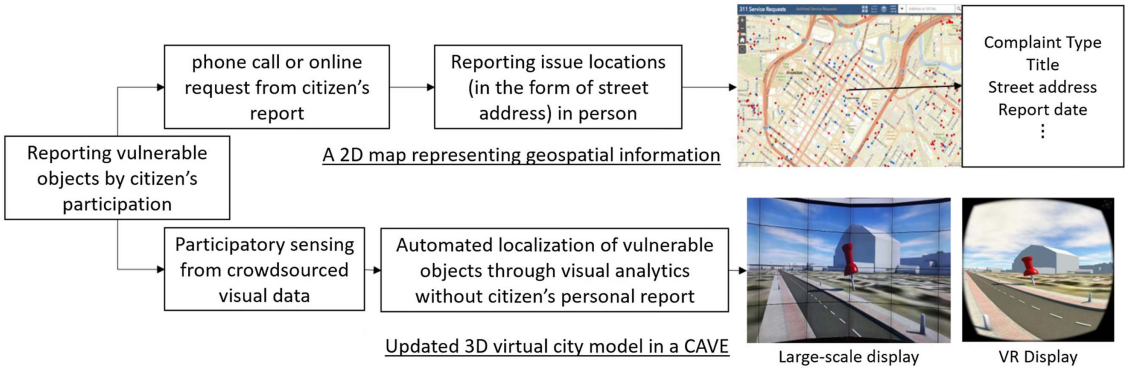


Fig. 7. Two different approaches for reporting and mapping vulnerable objects: (a) citizen reporting through 2D map-based phone call or online requests (e.g., Houston 311); and (b) the proposed method through participatory sensing and crowdsourced visual data analytics.

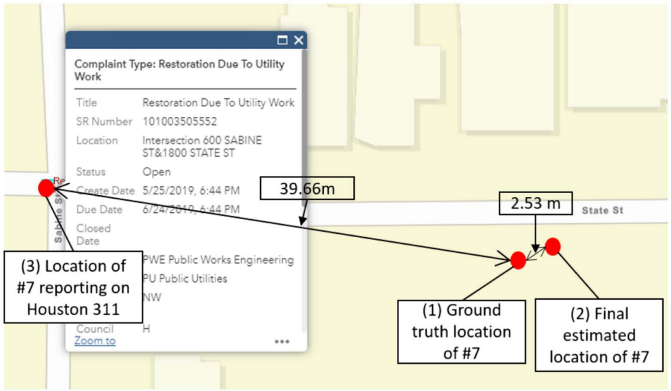


Fig. 8. Comparison of localization errors between the proposed method and Houston 311 service map for object #7: (1) ground truth location of the vulnerable object; (2) final estimated location of the object through the proposed method; and (3) mapping location of the object through Houston 311 service map.

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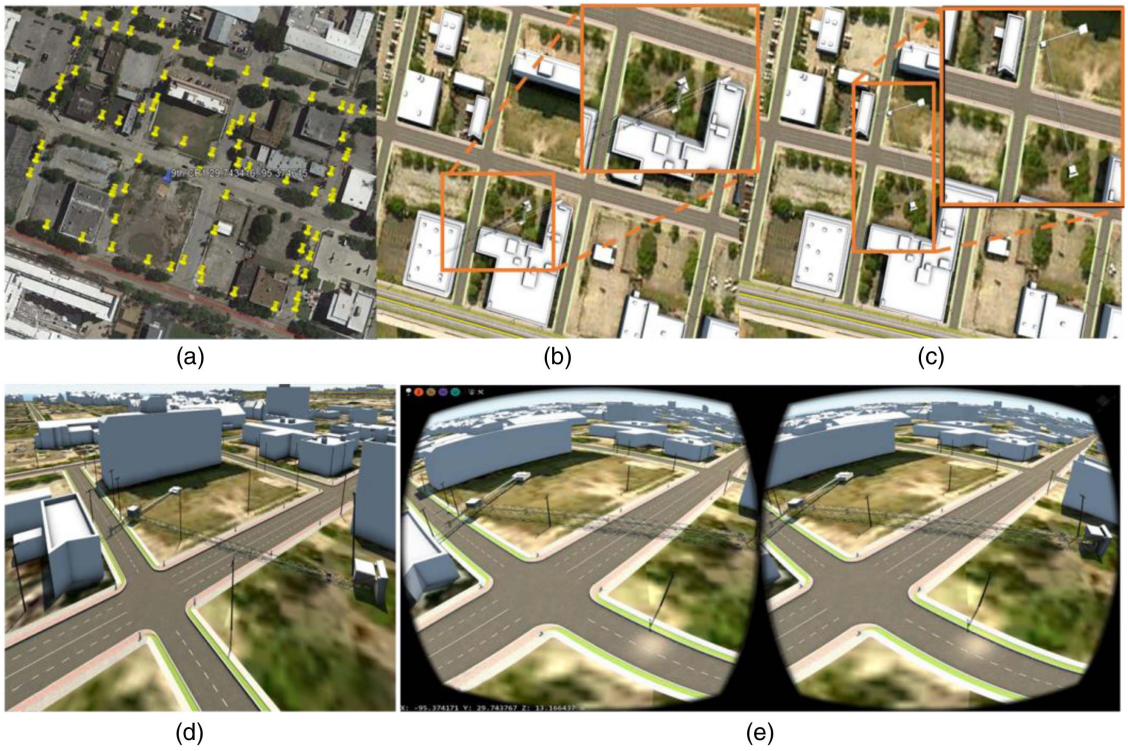


Fig. 9. Example scenario of a cascading effect on critical infrastructure (utility poles and metro rails) by failures of a vulnerable object (tower crane) in Houston. [(a) Imagery ©2019 Google, Map data ©2019.]

would threaten power cables directly connected with city power systems. In 2017, three high-rise cranes at South Florida construction sites collapsed as wind from Hurricane Irma intensified. The wildly swinging booms of the three cranes at buildings under construction bent or toppled over, sending debris onto city streets and damaging neighboring roads and power poles. There exists no readily available source of information about the as-is status of such vulnerable objects in cities.

In the proposed method, the geospatial information of such physical vulnerabilities and neighboring critical infrastructure can be explored in the participatory sensing–driven updated 3D virtual city model through the CAVE. This model update and the interactive visualization will allow city decision makers to predict secondary damage incurred by failures of vulnerable objects and further help analyze what-if scenarios with increased visibility into hazard and city interactions, which can support risk-informed decision-making for enhanced disaster preparedness and hazard mitigation. Fig. 9 shows an example scenario of cascading effects on neighboring critical infrastructure (utility poles and metro rails) by failures of a vulnerable object (tower crane) in Houston: (1) top view in Google Earth; (2) top view of a high-rise crane in a 3D city model; (3) top view of a crane collapse scenario in a 3D city model; (4) street view in the 3D city model; and (5) a VR display in the CAVE for interactive visualization.

Conclusion

This paper presents a new framework to build a digital twin city with participatory sensing to better understand spatiotemporal information of physical vulnerabilities for effective risk-informed decision-making. The proposed approach consists of (1) geospatial localization of vulnerable objects in the distance recognized from participatory sensing; and (2) a model update based on unstructured crowdsourced visual data analytics to better understand the spatiotemporal information of physical vulnerabilities with respect to neighboring critical infrastructure and interactive 3D visualization in the CAVE. To test the framework, case studies were conducted on Houston. The experimental results indicate that participatory sensing–based spatial information of vulnerable objects enables users to better understand the situational awareness of the extent to which given urban areas are vulnerable to anticipated disaster situations, as well as how such vulnerable objects likely affect neighboring critical infrastructure if they fail during extreme weather events.

Public opinion, such as risk perception of infrastructure regarding disaster situations, is increasingly used for associated infrastructure management (Valentin et al. 2018). For example, public awareness of flood risk on infrastructure could be a valuable insight to assist decision makers in establishing a city resilience strategy in the context of infrastructure disaster management (Al-Amin et al. 2019). In addition, after extreme weather events such as hurricanes, information from crowds regarding damaged infrastructure (e.g., geospatial information) can be used to facilitate transportation-related infrastructure management for disaster recovery (Dashti et al. 2014; Zhang et al. 2019). In this sense, research has been done on how to efficiently collect and analyze public opinion (Jiang et al. 2016; Tsujikawa et al. 2016; Zhang et al. 2019), and surveys have been most widely used for obtaining information from crowds about infrastructure reliability and risk awareness (Taeby and Zhang 2019; Tsujikawa et al. 2016). Despite the benefits of conventional questionnaires, there is a benefit from collecting and analyzing infrastructure-related data from crowds, as discussed in Kirilenko et al. (2015) and Kirilenko and Stepchenkova (2014), which can enhance data-driven infrastructure management. Here, the proposed digital twin city platform, building on crowdsourced visual data from participatory sensing, can be used as a support tool to collect and analyze infrastructure-related data

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from crowds in the context of disasters, and in turn positively support infrastructure disaster management.

Although this paper shows the potential of the proposed method for creating a digital twin city for effective risk-informed decision-making, open research challenges still remain. (1) The geospatial information of vulnerable objects from crowdsourced data would not be enough to build a digital twin city for effective risk-informed decision making: there is a need to better characterize as-is city conditions, such as infrastructure defects or neighborhood disorder, in a virtual environment to understand the extent to which given urban areas are susceptible to anticipated extreme weather events. Integrating various sources of information (e.g., textual/audio information from city dwellers regarding vulnerable objects or events, real-time data from existing IoT sensors) into the proposed digital twin city and the VR modeling platform would be a great way to reduce the gap between a real city and its digital replica, especially in the context of physical vulnerabilities in given areas, which can ultimately improve risk-informed decision making. (2) Participatory sensing can be powerful, by leveraging city dwellers' smartphones with mobility, and has significant potential to collect huge amounts of information at high spatiotemporal resolution. Despite the possible benefits, the success completely relies on the willingness and motivation of participants to conduct sensing tasks, and thus it is important to consider how we can ensure that city dwellers are willing to participate in data collection. One approach is to leverage crowdsourcing photos along with publicly available dataset such as Google Street view, as introduced in Cheng et al. (2018); Najafizadeh and Froehlich (2018); and Wang et al. (2018). (3) It is possible that the locations of vulnerable objects change infrequently, and such temporal variations also refer to the degree of changes in physical vulnerability associated with the vulnerable objects. In the proposed approach, it is technically possible that the localization of a vulnerable object can be dynamically updated based on additional visual data containing the same object. In this case, only the latest version of the localized vulnerability will be modeled and visualized in the virtual environment, as the old one will be replaced with the updated one. More case studies are needed to better evaluate the updating ability of the proposed digital twin city approach by taking into account spatiotemporal fluctuations of local vulnerabilities. Addressing these limitations and implementing future works are currently being explored as part of ongoing research.

Data Availability Statement

Some or all data, models, and code generated or used during the study are available from the corresponding author by request, including GPS and compass bearing data during experiments.

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